

CRISPR-X

Genome Engineering Lab • Interactive AI & Computational Biology Learning Platform

From concept to computational experiment. CRISPR-X turns complex genome-engineering workflows into an interactive browser experience—combining sequence editing, molecular analysis, drug-screening concepts, knowledge retrieval, and auditable experimentation in one educational environment.

THE PROBLEM

Genome engineering sits at the intersection of biology, computation and data science, but learning these workflows often requires disconnected tools, specialist software and access to laboratory or clinical infrastructure. This makes the underlying reasoning difficult to explore safely and interactively.

THE SOLUTION

CRISPR-X provides a self-contained, client-side sandbox where learners can explore simplified CRISPR-Cas9 editing, mutation detection, repair strategies, molecular visualization, computational compound screening, synthetic clinical-trial analytics, and a local CRISPR knowledge assistant.

WHAT MAKES IT DIFFERENT

Genome editing simulation	Sequence comparison, target selection, gRNA workflow, simulated Cas9 cutting and repair.
Computational chemistry	RDKit-derived molecular descriptors and 3D chemical-space exploration using synthetic compounds.
Biomedical AI learning	Real YOLOv8 vision inference plus a local TF-IDF/cosine retrieval assistant for CRISPR knowledge.
Auditable experimentation	Local SHA-256 hash-chain logging of important workflow events and synthetic trial actions.
Clinical analytics sandbox	Synthetic cohorts, adverse-event logging, Kaplan–Meier-style survival analysis and exportable data.

TECH STACK

Python • RDKit • JavaScript • HTML5 • CSS3 • Three.js • WebGL • TF-IDF • cosine similarity • YOLOv8 • SHA-256 • Kaplan–Meier statistics • synthetic data generation • browser-native APIs

WHAT'S NEXT

Move from educational simulation toward research-grade integrations: connect the audit ledger to the **Hedera** testnet/mainnet through the Hedera SDK; replace stylized molecular scoring with **AutoDock Vina** or **DiffDock**; and connect genomic analysis to **ClinVar**, **gnomAD** or **Ensembl** data sources.

Responsible by design: CRISPR-X is an educational demonstration, not a clinical, wet-lab or production system. It uses synthetic/illustrative data; no real patients or PII are used, and no external AI model, genome database, EHR system or live blockchain network is called by the current browser application.

PROJECT STATUS: COMPLETED • **FORMAT:** Single-file browser application • **MISSION:** Make advanced genome-engineering concepts more interactive, transparent and accessible.